D-SCRAMBLED CANTOR SETS

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Abstract. For all $d \geq 3$, we provide an example of a homeomorphism of Cantor space that has an uncountable d-scrambled set but no d-scrambled Cantor set.

Introduction

Suppose that X is a metric space and $f \colon X \to X$. We say that $x \in X^d$ is *proximal* if $\liminf_{n \to \infty} \max_{i<j<d} d_X(f^n(x(i)), f^n(x(j))) = 0$, *asymptotic* if $\limsup_{n \to \infty} \min_{i<j<d} d_X(f^n(x(i)), f^n(x(j))) = 0$, and *Li–Yorke* if it is proximal but not asymptotic. A set $Y \subseteq X$ is *d-scrambled* if every injective sequence in Y^d is Li–Yorke, and f is *Li–Yorke d-chaotic* if there is an uncountable d-scrambled set $Y \subseteq X$.

In [GGM21], it was shown that every Li–Yorke 2-chaotic continuous function on a Polish space has a 2-scrambled Cantor set. Here we show:

Theorem 1 (ZFC). *Suppose that $d \geq 3$. Then there is a Li–Yorke d-chaotic homeomorphism of $2^{\mathbb{N}}$ with no d-scrambled Cantor set.*

A *graph* on X is an irreflexive symmetric set $G \subseteq X \times X$. Let $\mathrm{Hyp}^d(G)$ be the d-ary relation on X of all injective sequences $x \in X^d$ such that $G \cap (x[d] \times x[d]) \neq \emptyset$. An *embedding* of a d-ary relation R on X into a d-ary relation S on Y is an injection $\pi \colon X \to Y$ such that $x \in R \iff \pi \circ x \in S$ for all $x \in X^d$. The *saturation* of a set $Y \subseteq X$ under a bijection $T \colon X \to X$ is given by $[Y]_T = \bigcup_{n \in \mathbb{Z}} T^n[Y]$. A subset of a topological space is $\mathbf{\Pi}^0_2$ if it is a countable intersection of open sets, and $\mathbf{\Sigma}^0_3$ if it is a countable union of $\mathbf{\Pi}^0_2$ sets. The primary new technical result underlying our proof of Theorem 1 is the following:

Theorem 2 (ZF + DC). *Suppose that $d \geq 3$ and G is a $\mathbf{\Sigma}^0_3$ graph on $2^{\mathbb{N}}$. Then there is a continuous embedding $\pi \colon 2^{\mathbb{N}} \to 2^{\mathbb{N}}$ of $\mathrm{Hyp}^d(G)$ into the d-ary Li–Yorkeness relation of a homeomorphism $T \colon 2^{\mathbb{N}} \to 2^{\mathbb{N}}$ for which there is a unique point of $2^{\mathbb{N}}$ not in the T-saturation of $\pi[2^{\mathbb{N}}]$.*

In §1, we establish Theorem 2. In §2, we obtain Theorem 1 and a related independence phenomenon as corollaries.

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1. Embedding hypergraphs

Define $S\colon 2^{\mathbb{Z}} \to 2^{\mathbb{Z}}$ by $S(c)(i) = c(i+1)$ for all $i \in \mathbb{Z}$ and $c \in 2^{\mathbb{Z}}$. Theorem 2 follows from [Kec95, Proposition 7.4] and:

Theorem 1.1 ($\mathtt{ZF}+\mathtt{DC}$). *Suppose that $d \geq 3$ and G is a $\mathbf{\Sigma}^0_3$ graph on $2^{\mathbb{N}}$. Then there exist an S-invariant perfect set $F \subseteq 2^{\mathbb{Z}}$, whose finite sequences are proximal, and a continuous embedding $\pi\colon 2^{\mathbb{N}} \to F$ of $Hyp^d(G)$ into the complement of the d-ary asymptoticity relation for which $0^{\mathbb{Z}}$ is the unique point of F not in the S-saturation of $\pi[2^{\mathbb{N}}]$.*

Proof. For all $s \in 2^{<\mathbb{N}}$, define $\mathcal{N}_s = \{c \in 2^{\mathbb{N}} \mid \forall i < |s|\ s(i) = c(i)\}$. For all $S \subseteq 2^{<\mathbb{N}}$, define $\mathcal{N}_S = \bigcup_{s \in S} \mathcal{N}_s$. For all $\ell \in \mathbb{N}$ and $p\colon 2^\ell \to d$, define $\ell_p = \ell$ and $\mathcal{N}_p = \prod_{i<d} \mathcal{N}_{p^{-1}(\{i\})}$. We say that p is a *near bijection* if it is surjective and there is a unique $k_p < d$ for which $|p^{-1}(\{k_p\})| > 1$. For all $k \in d \setminus \{k_p\}$, let $p^{-1}(k)$ denote the unique element of $p^{-1}(\{k\})$.

Fix $\mathbf{\Pi}^0_2$ graphs G_n on $2^{\mathbb{N}}$ for which $G = \bigcup_{n\in\mathbb{N}} G_n$, and decreasing sequences $(G_{m,n})_{m\in\mathbb{N}}$ of open graphs whose intersection is G_n. Set $H_{m,n} = \mathrm{Hyp}^d(G_{m,n})$ and $H_n = \bigcap_{m\in\mathbb{N}} H_{m,n}$. Then $\mathrm{Hyp}^d(G) = \bigcup_{n\in\mathbb{N}} H_n$. For all $m, n \in \mathbb{N}$, let $P_{m,n}$ denote the set of all near bijections p for which $\ell_p \geq m$, $p(0^{\ell_p}) = 0$, and $\mathcal{N}_p \subseteq H_{m,n}$. Let $Q_{m,n}$ denote the set of all $q \in P_{m,n}$ for which there is no $p \in P_{m,n}$ such that $k_p = k_q$, $\ell_p < \ell_q$, and $p^{-1}(k) = q^{-1}(k) \upharpoonright \ell_p$ for all $k \in d \setminus \{k_p\}$.

Fix an infinite set $I \subseteq \mathbb{N}\setminus\{0\}$ such that $|I \cap (I-n)| < \aleph_0$ for all $n > 0$, and a bijection $\phi\colon I \to \mathbb{N} \cup \bigcup_{m,n\in\mathbb{N}} \{m\} \times \{n\} \times Q_{m,n}$ with $I \cap N_i = \{i\}$ for all $i \in I \setminus \phi^{-1}(\mathbb{N})$, where $N_i = [i, 2i + (d-2)(\phi(i)(1)+1)]$. Define a continuous function $\pi\colon 2^{\mathbb{N}} \to 2^{\mathbb{Z}}$ by setting $\pi(c)(i) = 1$ if and only if

(1) $i = 0$;
(2) $i \in I$, $\phi(i) \in \mathbb{N}$, and $c(\phi(i)) = 1$; or
(3) $\exists 0 < k < d \exists m, n \in \mathbb{N} \exists q \in Q_{m,n}$
$((m,n,q) = \phi(i - (k-1)(n+1))$ and $q(c \upharpoonright \ell_q) = k)$.

Lemma 1.2. *The function π is injective.*

Proof. Suppose that $c, d \in 2^{\mathbb{N}}$ are distinct. Then there exists $n \in \mathbb{N}$ with $c(n) \neq d(n)$. Fix $i \in I$ for which $\phi(i) = n$. Then $\pi(b)(i) = b(n)$ for all $b \in \{c, d\}$, so $\pi(c)(i) \neq \pi(d)(i)$, thus $\pi(c) \neq \pi(d)$. $\square$

The *product metric* on $2^{\mathbb{Z}}$ is given by $\rho(c,d) = \sum_{i\in\mathbb{Z}} |c(i) - d(i)|/2^{|i|}$ for all $c, d \in 2^{\mathbb{Z}}$. A *homomorphism* from a d-ary relation R on a set Y to a d-ary relation S on a set Z is a function $\pi\colon Y \to Z$ such that $\pi \circ y \in S$ for all $y \in R$. Define $N_i = \{i\}$ for all $i \in \phi^{-1}(\mathbb{N})$.

Lemma 1.3. *The function π is a homomorphism from $Hyp^d(G)$ to the complement of the d-ary asymptoticity relation.*

Proof. Suppose that $c \in \mathrm{Hyp}^d(G)$, and fix $n \in \mathbb{N}$ and $j, k < d$ such that $c(j) \; G_n \; c(k)$. By permuting the coordinates of c, we can assume that either $c(0) = 0^{\mathbb{N}}$ or both $0^{\mathbb{N}} \notin c[d]$ and $j, k > 0$. Fix $k' \in d \setminus \{j, k\}$ if $c(0) = 0^{\mathbb{N}}$ and set $k' = 0$ otherwise. If $m \in \mathbb{N}$ is sufficiently large that $\mathrm{proj}_{2^m} \circ c$ is injective and $0^m \in (\mathrm{proj}_{2^m} \circ c)[d] \iff 0^{\mathbb{N}} \in c[d]$, then there is a least $\ell \geq m$ for which there is a nearly bijective extension $q \in P_{m,n}$ of $(\mathrm{proj}_{2^\ell} \circ c)^{-1}$ such that $k_q = k'$ and $\ell_q = \ell$. As $q \in Q_{m,n}$, there exists $i \in I$ with $\phi(i) = (m, n, q)$. Then $\mathrm{proj}_{2^{N_i}} \circ \pi \circ c$ is injective, so $\min_{j<k<d} \rho(S^i(\pi(c(j))), S^i(\pi(c(k)))) \geq 1/2^{(d-2)(n+1)}$. □

Lemma 1.4. *The function π is a homomorphism from $(2^{\mathbb{N}})^d \setminus Hyp^d(G)$ to the d-ary asymptoticity relation.*

Proof. Suppose that $c \in (2^{\mathbb{N}})^d$ and $\pi \circ c$ is not asymptotic. Fix $\epsilon > 0$ such that $I' = \{i' \in \mathbb{N} \mid \min_{j<k<d} \rho(S^{i'}(\pi(c(j))), S^{i'}(\pi(c(k)))) \geq \epsilon\}$ is infinite, as well as $r \in \mathbb{N}$ such that $\sum_{|i|>r} 1/2^{|i|} < \epsilon$. By throwing out finitely many elements of I', we can assume that for all $i' \in I$, there is at most one $i \in I$ such that $[i' - r, i' + r] \cap N_i \neq \emptyset$. Our choice of r then yields a unique such $i \in I$, in which case $\mathrm{proj}_{2^{N_i}} \circ \pi \circ c$ is injective. As $d \geq 3$, there exist $m, n \in \mathbb{N}$, $q \in Q_{m,n}$, and $\sigma \in \mathrm{Sym}(d)$ for which $(d-2)(n+1) \leq 2r$, $\phi(i) = (m, n, q)$, and $c \circ \sigma \in \mathcal{N}_q$. By passing to an infinite subset of I', we can assume that the values of k_q, n, and σ are independent of $i' \in I'$. The definition of $Q_{m,n}$ then ensures that elements of I' corresponding to the same value of m also correspond to the same value of i, so $c \in \bigcap_{m \in \mathbb{N}} H_{m,n} = H_n \subseteq \mathrm{Hyp}^d(G)$. □

The closure F of the S-saturation of $\pi[2^{\mathbb{N}}]$ is clearly perfect.

Lemma 1.5. *The sequence $0^{\mathbb{Z}}$ is the unique element of F not in $[\pi[2^{\mathbb{N}}]]_S$.*

Proof. Suppose that $c \in F \setminus [\pi[2^{\mathbb{N}}]]_S$. Then there exist a sequence $(c_n)_{n \in \mathbb{N}}$ of elements of $2^{\mathbb{N}}$, $\epsilon \in \{\pm 1\}$, and a strictly increasing sequence $(i_n)_{n \in \mathbb{N}}$ of natural numbers such that $S^{\epsilon i_n}(\pi(c_n)) \to c$. If $\epsilon = -1$, then clearly $c = 0^{\mathbb{Z}}$. If $\epsilon = 1$ and $c \neq 0^{\mathbb{Z}}$, then our choice of I ensures that c is in the S-orbit of $\pi(0^{\mathbb{N}})$, a contradiction. □

Lemma 1.5 implies that the union of the supports of finitely-many elements of F is contained in a finite union of translates of $\{0\} \cup \bigcup_{i \in I} N_i$. As the defining property of I ensures that such sets have arbitrarily long gaps, every finite sequence of elements of F is proximal. □

A set $Y \subseteq X$ is a *G-clique* if all distinct elements of Y are G-related, and *G-independent* if no elements of Y are G-related.

Proposition 1.6 ($\mathtt{ZF} + \mathtt{DC}$). *Suppose that $d \geq 2$, X and Y are Polish spaces, G is a Borel graph on X, $T \colon Y \to Y$ is a Borel automorphism,*

$B \subseteq Y$ is an uncountable Borel d-scrambled set, $\pi\colon X \to Y$ is an injective Borel homomorphism from $X^d \setminus Hyp^d(G)$ to the complement of the d-ary Li–Yorkeness relation, and the saturation of $\pi[X]$ is co-countable. Then G has a perfect clique.

Proof. Fix $i \in \mathbb{Z}$ for which $B \cap (T^i \circ \pi)[X]$ is uncountable. The perfect set theorem and Galvin's Theorem (see, for example, [Kec95, Theorems 13.6 and 19.7]) then yield a perfect set $P \subseteq (T^i \circ \pi)^{-1}(B)$ that is either a G-clique or G-independent. But G-independence would contradict the fact that B is d-scrambled. ☐

2. D-Scrambled sets

A subset of a topological space is $\mathbf{\Sigma^0_2}$ if its complement is $\mathbf{\Pi^0_2}$.

Proof (of Theorem 1). By [KV12, Theorem 2.1] and a simple Baire category argument, there is an $\mathbf{\Sigma^0_2}$ graph G on $2^{\mathbb{N}}$ with an uncountable clique $Y \subseteq 2^{\mathbb{N}}$ but no perfect clique. By Theorem 2, there is a continuous embedding $\pi\colon 2^{\mathbb{N}} \to 2^{\mathbb{N}}$ of $\mathrm{Hyp}^d(G)$ into the d-ary Li–Yorkeness relation of a homeomorphism $T\colon 2^{\mathbb{N}} \to 2^{\mathbb{N}}$ for which the saturation of $\pi[2^{\mathbb{N}}]$ is co-countable. Then $\pi[Y]$ is an uncountable d-scrambled set and there is no perfect d-scrambled set by Proposition 1.6. ☐

Let $\mathfrak{c}$ denote the cardinality of the continuum.

Theorem 2.1 ($\mathtt{ZFC} + \neg\mathtt{CH}$)**.** *Suppose that $d \geq 3$. Then it is consistent that there is a homeomorphism $T\colon 2^{\mathbb{N}} \to 2^{\mathbb{N}}$ with a d-scrambled set of cardinality $\mathfrak{c}$ but no perfect d-scrambled set.*

Proof. By [She99, Theorem 1.13], we can assume that there is a $\mathbf{\Sigma^0_2}$ graph G on $2^{\mathbb{N}}$ with a clique of cardinality $\mathfrak{c}$ but no perfect clique. The proof of Theorem 1 therefore yields the desired result. ☐

A *hypergraph* is a permutation-invariant set of injective sequences.

Theorem 2.2 ($\mathtt{ZFC} + \neg\mathtt{CH}$)**.** *Suppose that $d \geq 3$. Then it is consistent that every continuous function on a Polish space with a d-scrambled set of cardinality $\aleph_2$ has a d-scrambled perfect set.*

Proof. By [KS03, Proposition 3.4], we can assume that every Borel d-ary hypergraph on a Polish space with a clique of cardinality $\aleph_2$ has a perfect clique. But the d-ary Li–Yorkeness relation is Borel. ☐

Remark. For all $D \subseteq \mathbb{N} \setminus 3$, generalizations of our main results to individual sets that are simultaneously d-scrambled for all $d \in D$ can be obtained by interleaving the underlying arguments.

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